

# Wheels Docker Build — ccache & Performance Changes

## Problem

The `Dockerfile-Wheels-CIBW` build was slow because:

1. **No ccache** was configured inside the cibuildwheel-spawned containers, so the compilation of `oreanalytics_wrap.cpp` (a large SWIG-generated C++ file) happened from scratch for every Python version (cp310, cp312).
2. **SWIG generation ran redundantly** — `before_all_linux.sh` was set as `CIBW_BEFORE_BUILD`, which runs before *each* wheel, not once per platform.
3. **Dockerfile-Wheels-ORE** had ccache configured, but the `PATH` pointed to `/usr/lib/ccache` while `dnf`-based images place ccache symlinks in `/usr/lib64/ccache`, so ccache was not intercepting compiler calls via `PATH`.
4. **Both Dockerfiles shared the same BuildKit cache mount** (`--mount=type=cache,target=/ccache/`), but their CMake flags differed (e.g. `-fPIC`, different `-D` defines), so cached objects from one build never produced hits in the other.

## Changes

**ore/Docker/Dockerfile-Wheels-ORE**

| Change                                                                                  | Why                                                                                                             |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>PATH</code> updated to <code>/usr/lib64/ccache:/usr/lib/ccache:\$PATH</code>      | ccache symlinks are in <code>/usr/lib64/ccache</code> on <code>dnf</code> -based (RHEL/Fedora/manylinux) images |
| Cache mount changed to <code>--mount=type=cache,id=ccache-wheels,target=/ccache/</code> | Gives the wheels build its own dedicated BuildKit cache volume                                                  |

**ore/Docker/Dockerfile-ORE**

| Change                                                                               | Why                                                                           |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Cache mount changed to <code>--mount=type=cache,id=ccache-ore,target=/ccache/</code> | Explicitly names this cache volume so it stays separate from the wheels cache |

**ore/Docker/Dockerfile-Wheels-CIBW**

| Change                                                                                                     | Why                                                                                                                               |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Added <code># syntax = docker/dockerfile:1.2</code>                                                        | Enables BuildKit features                                                                                                         |
| Moved <code>before_all_linux.sh</code> from <code>CIBW_BEFORE_BUILD</code> to <code>CIBW_BEFORE_ALL</code> | SWIG wrapping ( <code>setup.py wrap</code> ) is Python-version-independent — now runs once per platform instead of once per wheel |
| Added <code>CIBW_BEFORE_BUILD="pip install setuptools"</code>                                              | Ensures <code>setuptools</code> is available in each per-Python-version venv for <code>setup.py build_ext</code>                  |
| Added <code>CIBW_ENVIRONMENT</code> with ccache config                                                     | Routes compiler calls through ccache inside cibuildwheel containers (see details below)                                           |

## CIBW\_ENVIRONMENT details

```
PATH=/usr/lib64/ccache:/usr/lib/ccache:$PATH
CCACHE_DIR=/project/.ccache
CCACHE_MAXSIZE=5G
```

- **PATH** — Prepends ccache symlink directories so compiler invocations from `setup.py build_ext` are intercepted by ccache. The `wheels_ore2` image already has ccache installed via `dnf`.
- **CCACHE\_DIR=/project/.ccache** — Uses the project mount directory (`/project`), which persists across Python versions within a single `cibuildwheel` run. This means `cp312` gets cache hits from `cp310`'s compilation of the same `oreanalytics_wrap.cpp`.
- **CCACHE\_MAXSIZE=5G** — Reasonable limit for the SWIG wrapper cache.

`ore/ORE-SWIG/Wheels-gitlab/before_all_linux.sh`

| Change                                                         | Why                                                                                              |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Removed <code>setuptools</code> from <code>pip3 install</code> | Now handled by <code>CIBW_BEFORE_ALL</code> and <code>CIBW_BEFORE_BUILD</code> in the Dockerfile |

## How ccache works in the pipeline

### Dockerfile-ORE (the main ORE build)

- Debian-based image, ccache symlinks at `/usr/lib/ccache`.
- CMake is configured with `-DCMAKE_CXX_COMPILER_LAUNCHER=ccache`.
- BuildKit cache mount `id=ccache-ore` persists the ccache directory across builds.
- On repeat builds with identical source, ccache produces direct hits.

### Dockerfile-Wheels-ORE (the wheels ORE library build)

- `dnf`-based `manylinux` image, ccache symlinks at `/usr/lib64/ccache`.
- CMake is configured with `-DCMAKE_CXX_COMPILER_LAUNCHER=ccache`.
- BuildKit cache mount `id=ccache-wheels` gives this build its own cache volume, separate from the main ORE build (different flags like `-fPIC`).
- On repeat builds with identical source and flags, ccache produces direct hits.

### Dockerfile-Wheels-CIBW (cibuildwheel — the Python wheel packaging step)

- Runs inside the `ore-build-dependencies` image.
- `cibuildwheel` launches its own Docker containers from the `wheels_ore2` image.
- Inside those containers, `setup.py build_ext` compiles `oreanalytics_wrap.cpp`.
- `CIBW_ENVIRONMENT` sets up ccache in those containers:
- Compiler calls go through ccache symlinks in `PATH`.
- Cache stored at `/project/.ccache` (shared across Python versions).
- **First Python version** (e.g. `cp310`): cache miss, full compilation.
- **Second Python version** (e.g. `cp312`): cache hit on the bulk of compilation (same `oreanalytics_wrap.cpp`, same ORE headers, same compiler flags).

## Expected speedup

| Area                                      | Before                            | After                                                         |
|-------------------------------------------|-----------------------------------|---------------------------------------------------------------|
| SWIG generation                           | Runs once per Python version (2×) | Runs once per platform (1×)                                   |
| oreanalytics_wrap.cpp compilation (cp310) | Full compile                      | Full compile (cold cache)                                     |
| oreanalytics_wrap.cpp compilation (cp312) | Full compile                      | ccache hit (fast)                                             |
| Subsequent CI runs (same source)          | Full compile                      | ccache hits in Dockerfile-Wheels-ORE via BuildKit cache mount |

## Cache isolation

| Dockerfile             | Cache mount id                    | Why separate                                                                                                 |
|------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Dockerfile-ORE         | ccache-ore                        | Flags include <code>-DORE_BUILD_SWIG=ON</code> , no <code>-fPIC</code> , <code>-DORE_ENABLE_OPENCL=ON</code> |
| Dockerfile-Wheels-ORE  | ccache-wheels                     | Flags include <code>-fPIC</code> , <code>-DORE_ENABLE_OPENCL=OFF</code> , tests disabled                     |
| Dockerfile-Wheels-CIBW | N/A (ephemeral / project/.ccache) | Only lives within a single cibuildwheel run; shares across Python versions                                   |

|  |  |
|--|--|
|  |  |
|--|--|

|  |  |
|--|--|
|  |  |
|--|--|

|  |  |
|--|--|
|  |  |
|--|--|